7.

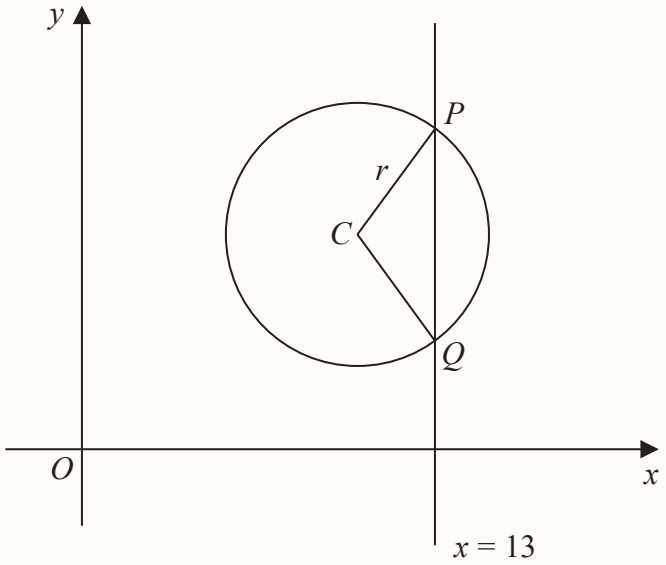


Figure 1

The circle with equation

$$x^2 + y^2 - 20x - 16y + 139 = 0$$

had centre C and radius r.

(a) Find the coordinates of C.

(2)

(b) Show that $r = 5$

(2)

The line with equation $x = 13$ crosses the circle at the points P and Q as shown in Figure 1.

(c) Find the y coordinate of P and the y coordinate of Q.

(3)

A tangent to the circle from O touches the circle at point X.

(d) Find, in surd form, the length OX.

(3)